**Question 1:** To be answered in (10) Minutes [ ] / 10 Marks

1. What is the closet time complexity in terms of **Big O** for the function $f(n) = n \log n + n^2$ ? **(0.5 marks)**

f(n) = $n^2$

2. If $f(n) = 2^{(3n)}$ and $g(n) = 2^n$, then write the relationships between f(n) and g(n) in terms of Ω. is: **(0.5 mark)**

$f(n) = \Omega\ g(n)$ $(2^3)^n$

f(n) greater than g(n)

3. For each of the following pairs of functions, determine which relationship is correct. **(3 marks)**

| | | f=O(g(n)) | f(n)=o(g(n)) | f(n)=Ω (g(n)) | f(n)=ω (g(n)) | f(n)= Θ (g(n)) |
|---|---|---|---|---|---|---|
| $f(n) = n^{\log m}$ | $g(n) = m^{\log n}$ | Yes | No | Yes | No | Yes |
| $f(n) = n^{(7/4)}$ | $g(n) = n \log_2(n)$ | Yes | Yes | No | No | No |

$n \log_2(n)$

4. Sort all the functions below in **increasing order** of asymptotic (big-O) growth.

$5^n$ , $n^{\log n}$ , 4 log n , $n^{1000}$

**(2 marks)**